Budget Front Desk

An agent built from a business's public pages, and the measurements of where it fails

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1. What was built

An AI agent for a real home-services company, built from a brief generated out of a ledger of 24 facts, each carrying the page and line it came from. The agent triages a complaint into emergency, diagnostic or free estimate, works the company's published price catalogue, collects what a dispatcher collects, and routes to a booking, a phone call or a payment.

The ledger bounds what goes into the brief: it fails the build if a price moves or if the evidence turns out to be template placeholder text. It does not bound what the model then says, which is why the output is checked separately against the same seven prices.

2. How it was measured

33 hand-written scenarios and 55 generated from the business's decision branches (service × county × timing × complication), three trials each. Every scenario carries required behaviours and forbidden ones; a forbidden behaviour fails it even when every required one is present. A deterministic grader checks the transcript, and a separate check fails any dollar figure that is not one of the seven published prices.

Two tiers are reported rather than one score: an **integrity gate** (any invented price, policy or placeholder fails the scenario outright) and **completeness** (how often the required behaviour appears), with a Wilson interval because the counts are small.

3. What the measurements found

	A0 as built	A2 two rules	A5 three rules
scenarios	53	53	55
trials	159	159	165
passed	113	135	145
strict pass[3]	26	40	45
integrity gate failures	0	0	0

Four instruction changes were tried. A1 regressed the emergency lane: told every reply must name a route, the agent chose "reply in chat" and dropped the phone number. A2 fixed that. A3 tried to make the agent state when the office reopens and it started querying a clock and contradicting a customer mid-gas-leak, so it was reverted. A5 reaches the same goal as A3 without a tool, and is the arm that ships.

By lane, on the scenarios both arms ran. The A2 column is included because it shows what A5 fixed and what it did not:

lane	A0	A2	A5
coverage	1/8	6/8	7/8
diagnostic	6/10	10/10	10/10
free estimate	6/8	8/8	8/8
emergency	4/8	6/8	6/8
trust	3/5	5/5	5/5
scheduling	3/9	2/9	4/9
off-task	2/3	1/3	2/3
handoff	1/2	2/2	1/2

4. What is wrong with it

- **One lane did not recover.** Handoff sits at 1/2: the multi-system and warranty-claim probes are where a person has to take over, and the agent handles one of the two.
- **Nineteen grader corrections** were needed, all found by reading transcripts rather than by the suite. Every one was a pattern failing a correct answer.
- **The comparison is significant only after those corrections.** The 27-scenario version of it was $p = 0.289$; it reached $p = 0.004$ after nineteen grader fixes, most of which were failing correct answers. That sequence is a reason to read the number with care rather than to celebrate it.
- **The held-out set was written after the first arm's failures were on screen.** It tests phrasing generalization, not failure modes nobody imagined.
- **One human calibration, 30 transcripts, and not double-blind.** Agreement was 70% with the requirement unstated and 97% with it stated; on the full 99-transcript set kappa fell to 0.314 because the sample is almost all passes. All three numbers say more about the rubric than about the agent.

5. What I would do next

1. Fix the handoff lane, then re-run the same set.
2. Move the three failure classes into the platform rather than the prompt: a typed next-step field, one clock resolved per turn with the customer's premise winning, and rule precedence as data. All three are implemented in `platform/` and replayed against the recorded conversations.
3. Get a second reader who never sees the first pass, on rows whose requirement is stated from the start.